Autonomous Engineering Platform (AEP) — Architecture Reference

System Architecture & Operational Tiers · **Version:** v1.2 (Immutable Artifact) · **Date:** 2026-11-09 · **Branch:** `docs/dual-lane-governance-proposal` · **Companion Docs:** `aep-hld-v1.2.html`, `aep-lld-v1.2.html`, `aep-backlog-v1.2.html`

This reference artifact captures the complete architectural topology of the Autonomous Engineering Platform (AEP) as the Agentic-Container OS. It formalizes two complementary operational tiers—fast inner-loop container lanes (Tier 1: Ralph) and sandboxed outer-factory pipelines (Tier 2: ESF)—governed by a shared substrate, fail-closed preflight gates, and cryptographic auditability.

1. High-Level Architecture Overview (AEP as the Agentic-Container OS)

End-to-end topology showing physical host hardware profiles, containment runtimes, control plane governance, substrate core services, dual-lane execution fabric, and verifiable settlement.

PHYSICAL HOST & RUNTIME SPECTRUM

Profiles: nano-8g (2vCPU/8GB N100) · small-16g (4vCPU/16GB) · medium-32g (8vCPU/32GB) · workstation-48g (12vCPU/48GB M3 Max)

macOS vz / Linux cgroups v

AEP CONTROL PLANE & GOVERNANCE LAYER

aep boot (Fail-Closed)
Preflight · Token Parity
Total RAM ≥ 7.5 GiB check

policy.data Guard
Git HEAD-Blob Cryptohash
Tamper lockout (Exit 1)

Dual Admission Control
FileLock (admission.lock)
Capacity Ceilings & Underflow-Safe RAM

Beads Ledger (SSOT)
bd list --limit 0 · DAG Blockers
tier:local vs tier:factory

Supervisor Dispatcher
§7 Deterministic Matrix
Quarantine-First (No Silent Downgrade)

AEP SUBSTRATE CORE SERVICES & IPC RELAYS (HOST RUNTIME)

NeuroStrata Cognitive Memory
ladybug.db · Hybrid Vector/FTS
Host Daemon (127.0.0.1:34343) / Compose aep-net

ghg Rate-Limited Gateway
3,500 req/h Cap · Coalesced Poller
Host GitHub Token Isolated (Zero Container Leak)

SUPERVISOR-MEDIATED IPC DOMAIN SOCKET RELAYS (/run/aep/lanes/<laneId>/)

memory.sock | ghg.sock | git.sock | exec.sock (Model)

TIER 1: FAST INNER LOOP (RALPH CONTAINER LANES)

Low/Standard Risk

- **Concurrency:** 1–2 lanes (Nano) up to 8–10 concurrent (Workstation-48g)
- **Execution Fabric:** OCI Container · Memory cap: 1.5–2 GB · tmpfs /tmp
- **Network Boundary:** **--network none (Zero Outbound Egress · No Leaked Keys)**
- **Git Isolation:** Isolated branch worktree · Private index · Host git.sock commit
- **Model Inference:** Host Harness executes model; sends tools over exec.sock
- **Verification:** Go-native aep verify · AST checks · Full-stream test logs

PARALLEL RALPH CONTAINERS (aep-lane-t1-xxxxxx)

Lane #1	Lane #2	Lane #3	Lane #N ...
Plan/TDD	Refactor	Testing	Workstation Max

TIER 2: OUTER FACTORY (ESF SANDBOX PIPELINES)

High Risk / Untrusted

- **Concurrency:** 1 pipeline (Nano) up to 4–6 concurrent (Workstation-48g)
- **Orchestration:** Temporal Worker · Replay-safe activities · esf-beads-<id>
- **Sandbox Hierarchy:** **Local-First (Cube/KVM → libkrun vz → gVisor systrap)**
- **Hardening & Probe:** ack_open_egress=false · In-VM EgressProbe SHA-256 evidence
- **Fallback Policy:** **Remote cluster is LAST RESORT · Quarantine-first on missing virt**
- **Re-Verification:** Fresh disposable sandbox meeting MinIsolation (Never on host)

PARALLEL ESF FACTORY PIPELINES (esf-beads-xxxxxx)

Pipeline #1	Pipeline #2	Pipeline #3	Remote Fallback
Cube/KVM microVM	libkrun (macOS vz)	gVisor systrap	Emergency Last Resort

CRYPTOGRAPHIC RECEIPT, AUDITABILITY & SETTLEMENT PIPELINE

Disposable Sandbox Re-Verification
Dynamic tests execute in fresh microVM meeting MinIsolation
Untrusted code never re-enters host kernel

Ed25519-Signed CommandReceipt
Full-stream StdoutSha + StderrSha + Merkle Tail + Nonce
Tamper-evident RFC 8785 canonical JSON

Ledger Settlement & Cognitive Inscription
Verified receipt required before bd close <issueId>
Continuous inscription into NeuroStrata memory

Dual-Constraint Admission
Simultaneously enforces hard concurrent lane count ceilings and cumulative RAM budgets parameterized by hardware profile. Protects host memory floor with underflow-safe arithmetic.

Zero-Egress Isolation
Lanes execute with `--network none`. All I/O occurs via host-side socket relays (`memory.sock`, `ghg.sock`, `git.sock`, `exec.sock`). Provider keys and PAT tokens never enter containers.

Local-First Sandbox Hierarchy
Tier 2 prioritizes local execution: (1) Linux KVM, (2) macOS `libkrun` (`vz`), (3) user-space `gVisor` (`systrap`). Remote clusters are strictly an emergency last resort; otherwise tasks safely quarantine.

Cryptographic Settlement
Command receipts bind full-stream stdout/stderr SHA-256 hashes and egress proofs signed with Ed25519. Required for Beads ledger closure and NeuroStrata memory inscription.

2. Dual-Lane Unified Architecture & ESF Deep Dive (Vertical Slices)

Detailed architectural decomposition comparing Tier 1 (Ralph Inner Loop) and Tier 2 (ESF Outer Factory), mapping the 7 vertical operational slices of the sandboxed factory.

TWO COMPLEMENTARY OPERATIONAL TIERS: INNER LOOP VS OUTER FACTORY

TIER 1: RALPH-LOOP (FAST INNER LOOP)

Workload Target: Features, refactors, tests, docs, low-blast-radius code
Isolation Level: OCI Container · cgroups v2 · Read-only rootfs · Private /tmp
Network Posture: **Zero direct egress (--network none) · IPC socket relays**
Inference Model: Host Harness executes model · Tools relayed via exec.sock
Git Workspace: Dedicated worktree branch · Private index · git.sock commit
Verification Gate: Go-native aep verify · Static analysis · Full test log capture

Speed: Seconds to minutes · Concurrency: 1–2 (Nano) to 8–10 (Workstation)

TIER 2: ESF PIPELINE (OUTER FACTORY)

Workload Target: Untrusted code/deps, CVE remediation, high-blast-radius execution
Isolation Level: **Hardware microVM (KVM/vz) or User-space kernel (gVisor)**
Network Posture: Measured closed boundary · Cryptographic in-VM EgressProbe
Orchestration: Temporal durable workflow · Replay-safe activities
Dependencies: Pre-baked base images · Read-only virtiofs volume caches
Re-Verification: Fresh disposable sandbox meeting MinIsolation (Never on host)

Depth: 15–45 min · Concurrency: 1 (Nano) to 4–6 (Workstation)

ESF (ENGINEERING SOFTWARE FACTORY) — CORE ARCHITECTURE & VERTICAL SLICES

Deterministic Sandboxed Lifecycle

SLICE 1: WORKFLOW

Temporal Engine

- Durable state
- Activity retries
- Replay safety
- Activity heartbeats

Idempotency Key
esf-beads-<id>
Prevents double-runs

Worker Pool
Queue: esf-pipeline
Max concurrent pollers
Profile-budgeted

Lifecycle States

1. Provision Sandbox
2. Ingest Repo Spec
3. Run Agent Engine
4. Probe Egress
5. Extract Patch
6. Re-Verify Sandbox
7. Sign Receipt & Settle

Failure Modes
Temporal lost: retry
Egress probe fail: halt
No silent downgrade

SLICE 2: SANDBOXES

Local-First Hierarchy

1. Cube / Linux KVM
Hardware microVM
/dev/kvm acceleration
Colima vz --nested-virt
Full kernel boundary

2. libkrun (macOS vz)
Apple Virtualization.fw
Direct hypervisor call
Zero nested KVM req
Hardware isolation

3. gVisor (runsc)
systrap mode (No KVM)
User-space kernel
Intel N100 / Cloud VMs
Syscall interception

Remote Sandbox
Emergency Last Resort
External Cube/E2B
Used ONLY if local fails
Explicit config req

Quarantine Invariant
If no backend valid:
quarantined_unsupported
Never auto-downgrade
Operator override only
Audited decision record

SLICE 3: AGENTS

Multi-Agent Engines

OpenCode v2
Autonomous codegen
Self-contained loop

Claude Code Engine
Multi-file refactorings
High reasoning tier

Codex / Pi Harness
Deterministic patches
Format compliance

Closed Dependency Cache

Pre-Baked Packages
Base image bakes compilers
Go, Rust, Node, Python

Read-Only Volume Cache
Shared package cache
Solves egress catch-22

Context Ingestion

- Target git commit SHA
- Beads issue requirements
- Injected test fixtures
- Invariant checklists

Prompt → Code → Patch

SLICE 4: EGRESS

factory.toml Guard

Hardening Policy
acknowledge_open =
false (Upstream Default)
allow_internet = false
Rejects open posture

in-VM EgressProbe

Active Network Probe
Fires DNS, TCP, ICMP
Asserts ENETUNREACH
Cryptographic Proof
SHA-256 evidence digest
Bound into run receipt

Enforcement vs Ack
PR Review Finding C1:
Ack flag only records risk;
does not enforce boundary.
AEP mandates active probe verification

Audit Verification
VerifyEgressProbe()
Validates probe signature
Corrupt/Missing probe →
Quarantine & Block PR

SLICE 5: IN-VM TEST

Deterministic Execution

Repo build.sh / test.sh
Runs in guest sandbox
Non-root user (aep:1000)
Captured exit codes
Full test log stream

Full-Stream Hashing
PR Review Finding C12:
Tails are display-only.
StdoutSha + StderrSha
Hashes every output byte
Detects silent failures

Cryptographic Patch
Diff extracted from guest
PatchDigest: SHA-256
Filtered for path traversal
Zero sensitive leak
Emitted as Artifact

Quality Gates

Uncle Bob (UB) Gates:

1. CRAP Score < 10
   CC > 8 forbidden
2. DRY Ratio < 0.8
   dupl AST scan
3. Mutation Kill > 85%
   100% boundary kill

SLICE 6: RE-VERIFY

Sandbox Re-Verification

Host Protection
PR Review Finding C4:
Running untrusted tests
in Tier 1 re-exposes
host to malicious code.
Host Kernel Protected

MinIsolation Rule
PR Review Finding C8:
Re-verification must match
or exceed run isolation.
gVisor runs → gVisor verify
KVM runs → KVM verify

Disposable Sandbox
Fresh instance spawned
Applies generated patch
Executes clean build/test
Destroyed on exit
Zero residual state

Tier 1 Static Gate
Tier 1 performs static
verification only:

- AST syntax check
- Diff cleanliness
- License scanning

SLICE 7: SETTLE

Audit & Settlement

Ed25519 Receipt
Canonical RFC 8785

- CommitSha
- Full StdoutSha
- EgressProof
- Nonce + LedgerSeq

Replay-protected

Beads Ledger Close
Command:
bd close <issueId>
Settlement Guard:
Requires verified receipt
Refuses unverified close
Single source of truth

Cognitive Inscription
Writes into NeuroStrata

- Root cause vector
- Fix diff embedding
- Edge case patterns
- Receipt Merkle hash

Continuous Learning

Auditor Check
Public key validation
Altered byte in test
fails verification
Non-repudiation
Audit Passed

> **ESF Vertical Slice Invariants:** Every ESF pipeline executes through all 7 vertical slices in order. An issue cannot settle (`bd close`) without passing in-VM egress proof verification (Slice 4), full-stream test logging (Slice 5), disposable sandbox re-verification meeting the original `IsolationClass` (Slice 6), and Ed25519 signature binding (Slice 7).
